Nakajima's representation of the Heisenberg algebra

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Plan of the talk

Heisenberg Algebra

Hilbert Scheme of Points

Borel-Moore Homology

Correspondences

Main Construction

Idea of Proof

Heisenberg algebra

The Heisenberg algebra is $\hat{\mathfrak{a}} = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}h_n \oplus \mathbb{C}K$ with bracket given by

$$[h_m, h_n] = m\delta_{m,-n}K, \qquad [K, \hat{\mathfrak{a}}] = 0,$$

and highest weight representation

$$H = \mathbb{C}[x_1, x_2, \ldots]$$

with the action

$$h_n \mapsto \begin{cases} n\frac{\partial}{\partial x_n} & \text{if } n > 0 \\ x_{-n} & \text{if } n < 0 \\ 0 & \text{if } n = 0 \end{cases}, \qquad K \mapsto \mathrm{Id}.$$

Hilbert scheme of points

Let X be a projective surface over $\mathbb{C}$. The essential property of the *Hilbert scheme of n points* $X^{[n]}$ is that a point (over $\mathbb{C}$) of $X^{[n]}$ is a 0-dimensional closed subscheme of X of length n.

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This **does not** mean that the points in $X^{[n]}$ are n-tuples of points of X counted with multiplicities.

Hilbert scheme of points

Theorem (Fogarty)

If X is a nonsingular projective surface,

1. *$X^{[n]}$ is nonsingular and has dimension $2n$, and*
2. *the Hilbert-Chow morphism*

$$\begin{aligned} \pi : X^{[n]} &\longrightarrow S^n X \\ Z &\longmapsto \sum_{x \in suppZ} \dim \Gamma(\mathcal{O}_{Z,x})[x] \end{aligned}$$

is a resolution of singularities.

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3. (Intersection pairing.) If X and Y are both "nicely embedded" in the same manifold M, there is a map

$$\cap_M : H_i^{BM}(X) \times H_j^{BM}(Y) \to H_{i+j-m}^{BM}(X \cap Y).$$

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4. (Fundamental class.) If X is a variety of dimension d over $\mathbb{C}$ then there is *fundamental class* $[X] \in H_{2d}^{BM}(X)$.

Correspondences

Let X_1, X_2 be smooth irreducible varieties of dimensions d_1, d_2. Let Z be a d-dimensional closed subvariety of $X_1 \times X_2$ such that the projections p_1 and p_2 are proper.

The *correspondence* associated to $Z \subset X_1 \times X_2$ is the map

$$\begin{aligned} c_{[Z]} : H_i^{BM}(X_2) &\longrightarrow H_{i+2d-2d_2}^{BM}(X_1) \\ c &\longmapsto p_{1,*}(p_2^* c \cap [Z]) \end{aligned}$$

where p_1 and p_2 are the projections in the following diagram:

$$\begin{tikzcd} & Z \subset X_1 \times X_2 \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X_1 & & X_2. \end{tikzcd}$$

Composition of correspondences

Let $Z_{12} \subset X_1 \times X_2$ and $Z_{23} \subset X_2 \times X_3$ such that the projections to X_2 and X_3 are proper. The composition of correspondences is

$$[Z_{12}] \circ [Z_{23}] := p_{13,*}(p_{12}^*[Z_{12}] \cap p_{23}^*[Z_{23}]),$$

$$\begin{tikzcd}
 & X_1 \times X_2 \times X_3 \arrow[dl, "p_{12}"'] \arrow[dr, "p_{23}"] \arrow[dd, "p_{13}"'] & \\
X_1 \times X_2 & & X_2 \times X_3 \\
 & X_1 \times X_3. &
\end{tikzcd}$$

In fact, the projection $p_{13}(p_{12}^{-1} Z_{12} \cap p_{23}^{-1} Z_{23}) \to X_3$ is proper.

Main construction

Let $i > 0$. Define $P[i] \subseteq \coprod_n X^{[n-i]} \times X^{[n]} \times X$ by

$$P[i] := \coprod_n \{(\mathcal{J}_1 \supseteq \mathcal{J}_2, x) : \mathrm{supp}(\mathcal{J}_1/\mathcal{J}_2) = \{x\}\}.$$

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Let $\alpha \in H_*^{BM}(X)$, $\beta \in H_*(X)$ and $i > 0$. Define

$$P_\alpha[i] = \varpi_*(\Pi_1^*\alpha \cap [P[i]])$$
$$P_\beta[i] = \varpi_*(\Pi_1^*\beta \cap [P[-i]]).$$

$$\begin{tikzcd} & \coprod_n X^{[n-i]} \times X^{[n]} \times X \arrow[dl, "\Pi_1"'] \arrow[dr, "\varpi_i"] & \\ X & & \coprod_n X^{[n-i]} \times X^{[n]}. \end{tikzcd}$$

Main theorem

Theorem (Nakajima, Grojnowski)

If $(deg\alpha)(deg\beta) \equiv 0 \ mod\ 2$*,*

$$[P_\alpha[i], P_\beta[j]] = (-1)^{i-1} i\delta_{i,-j} \langle \alpha, \beta \rangle Id,$$

where $\langle \alpha, \beta \rangle = p_*(\alpha \cap \beta)$ *for* $p : X \to pt$ *the unique morphism from* X *to the point* pt*.*

Easy case

Let $x \in X$, $\alpha = [X]$, $\beta = [x]$. Take a set of distinct n points $x_1, \ldots, x_n \in X \setminus \{x\}$ and identify them with the ideal $\mathcal{J}$. Then

$$[P_\alpha[1], P_\beta[-1]][\mathcal{J}] = [\mathcal{J}].$$

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$$\begin{tikzcd} & X^{[n+1]} \times X^{[n]} \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X^{[n+1]} & & X^{[n]}. \end{tikzcd}$$

- $P_{[x]}[-1][\mathcal{J}] = [\{x\} \cup \mathcal{J}]$, since we are looking at

$$\{(\mathcal{J}' \supseteq \mathcal{J}, x) : \mathrm{supp}(\mathcal{J}'/\mathcal{J}) = \{x\}\} \subset X^{[n]} \times X^{[n+1]} \times X.$$

- $P_{[X]}[1]P_{[x]}[-1][\mathcal{J}]$, since

$$\{(\mathcal{J}'' \subseteq \mathcal{J}', x) : \mathrm{supp}(\mathcal{J}'/\mathcal{J}'') = \{y\},\ \text{for some } y \in X\}$$

Easy case

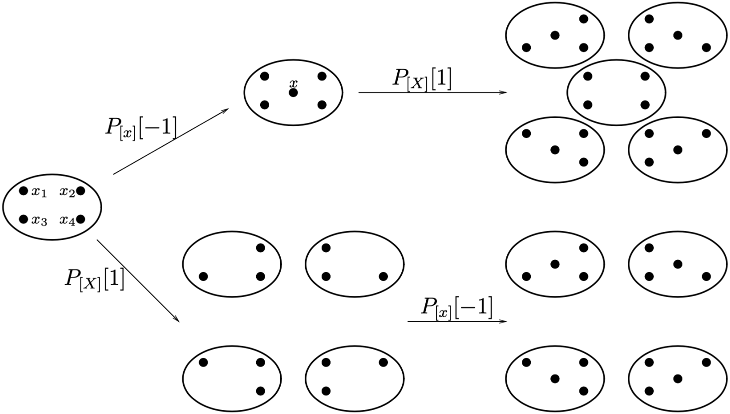

FIGURE 8.1. The proof of $\left[P_{[X]}[1], P_{[x]}[-1]\right][\mathcal{J}_1] = [\mathcal{J}_1]$

Proof for $i, j > 0$

Let's explain the case $i, j > 0$. By definition,

$$P_\alpha[i]P_\beta[j] = p_{13,*}(p_{12}^*[P[i]] \cap \Pi_1^*\alpha \cap p_{23}^*[P[j]] \cap \Pi_2^*\beta).$$

$$\begin{tikzcd}
 & X^{[n-i-j]} \times X^{[n-j]} \times X^{[n]} \arrow[dl, "p_{12}"'] \arrow[dd, "p_{13}"'] \arrow[dr, "p_{23}"] & \\
P[j] \subset X^{[n-j]} \times X^{[n]} & & X^{[n-i-j]} \times X^{[n-j]} \supset P[i] \\
 & X^{[n-i-j]} \times X^{[n]} &
\end{tikzcd}$$

Proof for $i, j > 0$

$$P[i,j] := p_{12}^{-1}(P[i]) \cap p_{23}^{-1}(P[j]) \cap (X^{[n-i-j]} \times X^{[n-j]} \times X^{[n]})$$

$$= \left\{ (\mathcal{J}_1 \supseteq \mathcal{J}_2 \supseteq \mathcal{J}_3) : {\substack{\mathrm{supp}(\mathcal{J}_1/\mathcal{J}_2)=\{x\} \text{ for some } x\in X \\ \mathrm{supp}(\mathcal{J}_2/\mathcal{J}_3)=\{y\} \text{ for some } y\in X}} \right\}$$

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$P[i,j]$ is the disjoint union of of $\{x \neq y\}$ and $\{x = y\}$.

Let $\overline{P}[i,j] := \overline{P[i,j] \cap \{x \neq y\}}$. Then

$$p_{12}^*[P[i]] \cap p_{23}^*[P[j]] = [\overline{P}[i,j]] + \iota_* R$$

where $R \in H_*^{BM}(P[i,j] \cap \{x = y\})$ corresponds to the $\{x = y\}$ part of $P[i,j]$ and ι is the inclusion.

Proof for $i, j > 0$

Lemma

1. *There exists an isomorphism*

$$P[i,j] \cap \{x \neq y\} \xrightarrow{\cong} P[j,i] \cap \{x \neq y\},$$

which exchanges Π_1 *and* Π_2*.*

2. $p_{13,*}(\Pi_1^* \alpha \cap \Pi_2^* \beta \cap \iota_* R) = 0, \qquad p_{13,*}(\Pi_1 \beta \cap \Pi_2^* \alpha \cap \iota_* R') = 0.$

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This lemma implies the result since

$$\begin{aligned} P_\alpha[i]P_\beta[j] &= p_{13,*}(\overline{P}[i,j] \cap \Pi_1^*\alpha \cap \Pi_2^*\beta) \\ &= (-1)^{\deg\alpha \deg\beta} p_{13,*}(\overline{P}[j,i] \cap \Pi_1^*\beta \cap \Pi_2^*\alpha) \\ &= P_\beta[j]P_\alpha[i]. \end{aligned}$$

Proof of 1.

Define an isomorphism by

$$\mathcal{J}_1/\mathcal{J}_2' = \mathcal{J}_2/\mathcal{J}_3, \qquad \mathcal{J}_2'/\mathcal{J}_3 = \mathcal{J}_1/\mathcal{J}_2.$$

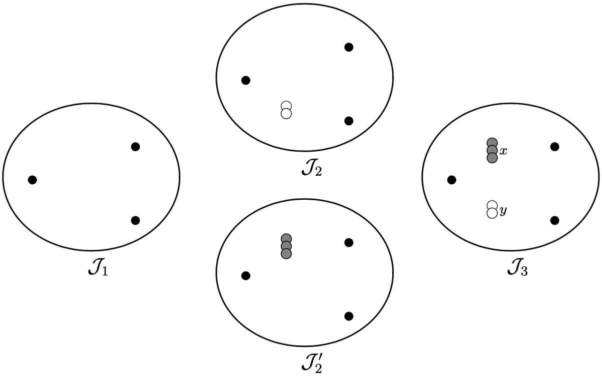

FIGURE 8.2. Correspondence between $\mathcal{J}_1 \supset \mathcal{J}_2 \supset \mathcal{J}_3$ and $\mathcal{J}_1 \supset \mathcal{J}_2' \supset \mathcal{J}_3$

Proof of 2.

Consider the commutative diagram

$$\begin{tikzcd}
P[i,j] \cap \{x = y\} \arrow[r, "\Pi'"] \arrow[d, "\iota"'] & X \arrow[d, "\Delta"] \\
P[i,j] \arrow[r, "\Pi_1 \times \Pi_2"'] & X \times X.
\end{tikzcd}$$

where Δ is the diagonal embedding and Π' is any of the projections Π_1 or Π_2. Then

$$\begin{aligned}
\Pi_1^*\alpha \cap \Pi_2^*\beta \cap \iota_* R &= \iota_*(\iota^*(\Pi_1^*\alpha \cap \Pi_2^*\beta) \cap R) \\
&= \iota_*(\iota^*(\Pi_1 \times \Pi_2)^*(\alpha \times \beta) \cap R) \\
&= \iota_*(\Pi'^*\Delta^*(\alpha \times \beta) \cap R) \\
&= \iota_*(\Pi'^*(\alpha \cap \beta) \cap R).
\end{aligned}$$

Proof of 2.

Consider another commutative diagram:

$$\begin{tikzcd} P[i,j] \cap \{x=y\} \arrow[rr, "\Pi'"] \arrow[d, "p_{13} \circ \iota"'] & & X \arrow[d, equal] \\ P[i+j] = \{(\mathcal{J}_1 \supseteq \mathcal{J}_3) : \mathrm{supp}(\mathcal{J}_1/\mathcal{J}_3) = \{x\}\} \arrow[rr, "\Pi''"'] & & X, \end{tikzcd}$$

where Π'' is the usual projection to X from the set $P[i+j]$. Then

$$p_{13,*}\iota_*(\Pi'^*(\alpha \cap \beta) \cap R) = \Pi''^*(\alpha \cap \beta) \cap p_{13,*}\iota_*(R).$$

Since $P[i+j]$ has less than the "expected dimension", $p_{13,*}\iota_*(R) = 0$.

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